

The Lensing Trilemma for Modified Inertia

Why a dark-matter-free acceleration-scale theory cannot bend light with pure modified inertia on one metric — and the two curvature-touching roads that remain: an elastic dark-energy medium, or Deffayet–Woodard nonlocal gravity

Carl P. Zimmerman · Briar Creek Tech · 2026-07-11 (v2 — closes the one loophole v1 left open: horn C's anisotropic-stress/slip channel, by a support-plus-multipole argument, §2.1)

Abstract

The de Sitter–Unruh program reads the MOND scale as a modified-**inertia** effect, $a_0 = c^2 \sqrt{\Lambda/32\pi} = cH_\Lambda/Z$. Modified inertia (MI) evades the solar-system tests that exclude modified-gravity MOND, but it has a standing gap: light bending. This note settles the structural question — *is there a pure-MI channel that bends light with the correct ν -enhancement using one metric (so it is GW170817-safe) and no dark component?* — and the answer is a clean **trilemma theorem: there is not**. To place the ν -enhancement into light deflection, the photon must couple to whatever carries ν , and there are exactly three homes for it. **(A)** The single shared metric felt by photons *and* gravitons: this lenses correctly and has $c_{\text{GW}} = c$ exactly (GW170817-safe), but a massive body is then a geodesic of that same enhanced metric, so its modified-inertia response must switch off ($\mu \rightarrow 1$) or galaxy rotation curves over-predict by ν — i.e. this is modified **gravity**, not modified inertia. **(B)** A species-split second metric (a disformal photon cone): the photon and graviton cones differ unless the disformal coefficient vanishes, and GW170817 forces it to vanish — which is identically the no-lensing corner (in the conformal limit, null cones are conformally invariant in four dimensions and light bends by *exactly zero*). **(C)** Modification confined to the matter/inertia sector — the framework's actual channel: the gravitational field equation is left baryonic (or, if the modified matter stress-energy were large enough to lens, a geodesic body would feel it and rotation curves would over-predict by ν unless $\mu \rightarrow 1$, collapsing to horn A). Options A, B, C exhaust the possibilities consistent with the equivalence principle and a single null cone; there is no fourth geometric home. The consequence is **not** that the program needs dark matter — it does not, and every surviving road is dark-matter-free. It is that correct lensing **requires touching curvature**, via exactly one of two roads: **(1)** an extra gravitating stress-energy that tracks the baryons — the framework's own **elastic dark-energy medium** (Branch B), which sources lensing at the cost of a marginal Cassini quadrupole; or **(2)** a genuinely distinct third option surfaced here, **Deffayet–Woodard nonlocal modified gravity**, which lenses correctly from one metric, is GW170817-safe by construction, and can host a_0 and ν — at the honest price that it is modified gravity (curvature-side nonlocality, geodesic matter), treats a_0 as a free input rather than deriving 9.36×10^{-11} , and has an as-yet-unproven quadrupole evasion. The pure-MI, one-cone, medium-free lensing channel — the single cleanest form of the no-dark-matter claim — is eliminated. A magnitude sub-claim from earlier internal notes, that pure MI under-lenses by 10^7 , is **withdrawn** as a dimensional error: the a_0 lensing potential is in fact $2.7\text{--}245 \times$ *larger* than the required extra potential, so the deficit is structural (which sector the enhancement lives in), not one of magnitude.

1. The gap, stated precisely

Modified inertia changes how a massive body responds to a given gravitational field: its equation of motion is $\mu(a/a_0) a = g_N$ with $\mu \rightarrow 1$ at high acceleration and $\mu \rightarrow a/a_0$ deep in the MOND regime, reproducing the flat rotation curve $v^4 = GMa_0$. Photons are massless; they follow null geodesics of the metric $g_{\mu\nu}$, which solves $G_{\mu\nu} = 8\pi G T_{\mu\nu}$. If the gravitational source is the baryonic $T_{00} = \rho_{\text{bar}}$, light is deflected as in unmodified Newtonian gravity — under-lensed, relative to the dynamics, by exactly the factor $\nu = g_{\text{obs}}/g_{\text{bar}}$. The observed weak-lensing radial-acceleration relation, however, is measured to *equal* the dynamical one out to $\sim\text{Mpc}$ scales [Brouwer et al. 2021; Mistele & McGaugh 2024]. So a viable dark-matter-free theory must bend light by νg_{bar} , not g_{bar} . The question of this note is whether pure MI — with a single metric, so that GW170817 is respected — can do that.

2. The trilemma

For the enhancement factor ν to appear in the deflection of light, the photon must couple to whatever object carries ν . There are exactly three such objects.

Horn A — the shared metric/connection. Put ν into the one metric that photons and gravitons both propagate on. Then $c_{\text{GW}} = c$ exactly and GW170817 is satisfied, and light lenses correctly. But a massive test body is a geodesic of that same metric, so it already experiences νg_{bar} ; for the rotation curve to come out right, the modified-inertia response must then be *absent* ($\mu \rightarrow 1$), because the dynamics fix $a_{\text{obs}} = \nu g_{\text{bar}}$ exactly once. A partial split — the metric carries ν^a and the inertia ν^{1-a} — under-lenses for any $a < 1$, since the photon receives only ν^a ; correct lensing forces $a = 1$, i.e. all of the enhancement in the metric and none in the inertia. This is **modified gravity**. It is realized concretely by Deffayet-Woodard nonlocal metric MOND [Deffayet, Esposito-Farèse & Woodard 2011] and by $f(Q)/f(T)$ MOND [D’Ambrosio, Garg & Heisenberg 2020], in which matter is minimally coupled and geodesic while the MOND force and the lensing enhancement both come from the modified geometric (left-hand) side.

Horn B — a second, species-split cone. Give photons their own metric $\tilde{g}_{\mu\nu} = g_{\mu\nu} + B u_\mu u_\nu$ (disformal). This can bend light correctly, but the photon and graviton cones then differ, producing a photon-versus-graviton arrival-time offset. GW170817 constrains that offset to $|\Delta c|/c \lesssim 10^{-15}$, which forces $B \rightarrow 0$ — and $B = 0$ is identically the no-lensing corner. In the purely conformal sub-case $\tilde{g} = \Omega^2 g$, the obstruction is exact and geometric: null cones are conformally invariant in four dimensions, so a conformal factor bends light by *exactly zero*, whatever its profile. This horn is **dead** by GW170817 (the offset for the framework’s disformal profile is 10^{5-6} over the bound).

Horn C — matter/inertia only. Confine the modification to the matter sector, as genuine modified inertia does (the framework’s nonlocal operator $K(\square_u/a_0^2)$ acts on the matter kinetic term, not on curvature). Then the gravitational field equation retains its baryonic source and the metric potential stays baryonic, so light under-lenses. One might hope the modified matter stress-energy $T_{\mu\nu}$ itself supplies the enhancement; but if it did — if T_{00} were enhanced to $\nu \rho_{\text{bar}}$ — a massive body would feel that enhanced field too, and rotation curves would over-predict by ν unless the inertia response switched off, which is precisely horn A again. So horn C either under-lenses or collapses into horn A.

Exhaustiveness. Any way light can acquire the ν -enhancement is a choice of what the photon couples to: the shared metric (A), a private metric (B), or nothing gravitational beyond the baryonic source (C). Under the equivalence principle and a single null cone, these exhaust the possibilities. Hence: **no pure modified-inertia theory — one cone, no medium, no dark component — bends light with**

the correct ν -enhancement. Correct lensing requires either an added gravitating stress-energy (a medium) or an enhanced metric (modified gravity).

2.1 Closing horn C's one loophole: the anisotropic-stress (slip) channel

Horn C as stated assumes the modified-inertia matter stress-energy is effectively isotropic, so that the two metric potentials are equal, $\Psi = \Phi$, and light (which is deflected by $\Psi + \Phi$) feels the same potential as the dynamics (governed by Φ). One loophole remains: an *anisotropic* stress sources a gravitational slip $\Psi \neq \Phi$, and in principle a slip could enhance $\Psi + \Phi$ (lensing) while leaving Φ (rotation curves) untouched – a correct-lensing channel with no double-count. We close this loophole by direct computation of $T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \delta S_{\text{matter}} / \delta g^{\mu\nu}$ for the framework's nonlocal action, keeping the derivative-of-operator (K') terms.

Two structural facts close it, both independent of the delicate magnitude bookkeeping and both a_0 -footing-independent. **(i) The source is baryon-confined.** Every term of the variation carries an explicit factor of the matter density ρ_m : the nonlocality lives in the *argument* of $K(\square_u/a_0^2)$, not in the support of the stress. So T_{00} vanishes identically wherever $\rho_m = 0$ – precisely the extended halo region where lensing measures the “missing” mass. A matter action cannot build a vacuum-supported dark source; that is the function of a modified-gravity field equation, of which there is none in horn C. (A supporting exact result: for matter on circular orbits the operator argument is fixed by the identity $u^\mu \square_u u_\mu = -a_\mu a^\mu = -a_c^2$, so $\square_u/a_0^2 = (a_c/a_0)^2 = O(1)$ in the MOND regime – the naive small orbital-frequency does *not* enter, and with $\|K\| \leq 1$ the energy density is $T_{00} \leq \rho_{\text{bar}}$, mildly *reduced*, never enhanced to $\nu \rho_{\text{bar}}$, so horn C does not secretly collapse into horn A.) **(ii) The slip is a compact traceless quadrupole.** The slip is sourced by the traceless anisotropic stress Π_{ij} ; for any compact traceless source the monopole of $\Psi - \Phi$ vanishes identically (two applications of the divergence theorem), so the far-field slip decays as $1/r^3$. Correct lensing instead requires $\Psi + \Phi$ to grow as $+\log r$ (enclosed lensing mass $\propto r$). A decaying quadrupole cannot furnish a rising logarithm. Granting even the most generous local anisotropic stress – of order $\rho_m c^2$ on the baryons – leaves it baryon-confined and traceless-compact, so it still cannot lens. The loophole is closed.

(An earlier version of this argument closed the loophole by an order-of-magnitude estimate, $\Psi - \Phi \sim (v/c)^2 \Phi$, a factor $\sim 10^6$ short. That estimate dropped the K' acceleration term and is not the correct closure; the support-plus-multipole argument above does not depend on the disputed ρc^2 -versus- ρv^2 magnitude and is the honest reason horn C holds.)

3. The two roads that survive – both dark-matter-free

The theorem eliminates the pure-MI channel but leaves the no-dark-matter program two genuine roads, because it forces the enhancement to touch curvature and both ways of doing so keep dark matter out.

Road 1 – the elastic dark-energy medium (Branch B). Let the dark energy itself, an elastic solid whose relaxed state is the cosmological constant, deform in the baryonic field and gravitate as a real source $T_{00} \rightarrow \nu \rho_{\text{bar}}$ [companion note, this program]. This is *not* dark matter – it is the dark energy already in the model, strained – and it lenses correctly on one metric (GW170817-safe by construction). Its cost is a Cassini quadrupole that this program has shown reduces to a single free material scalar (the shear Poisson ratio) whose natural values marginally fail, leaving the branch formally open but evidence-tilted.

Road 2 — Deffayet–Woodard nonlocal gravity. A purely metric theory with a nonlocal term (functions of $\square^{-1}$ acting on curvature) that reproduces MOND dynamics *and* correct weak lensing from one metric [Deffayet, Esposito-Farèse & Woodard 2011; and a 2026 single-interpolating-function realization]. It is GW170817-safe by construction (the MOND term carries two Ricci factors and vanishes for tensor radiation), and its solar-system screening is *exponential* in the field, $f(Z) \sim Z e^{-\frac{1}{3}\sqrt{|\bar{Z}|}}$ — structurally stronger than the power-law screening of the vector modified-gravity class, so it may evade the Cassini quadrupole better than Road 1. Its honest costs: it is modified **gravity**, not inertia, so it abandons the program’s defining trajectory-dependent-inertia identity; it takes a_0 as a free observational input and does *not* derive 9.36×10^{-11} ; and only its scalar post-Newtonian screening has been demonstrated — the anisotropic quadrupole is unverified.

The lensing decision is therefore a **trichotomy** — Road 1, Road 2, or abandon the no-dark-matter premise — and the third is the one this program rejects. Pure-MI-does-everything (the eliminated leg) is gone; the two survivors are both dark-matter-free.

4. A withdrawn magnitude claim, and what is genuinely open

Earlier internal notes stated that the pure-MI matter stress-energy is 10^7 too weak to source the lensing halo. That specific number is **withdrawn** as a dimensional error: the acceleration-scale potential $a_0 r / c^2$ is in fact $2.7 \times$ (at 5 kpc) to $245 \times$ (at 300 kpc) *larger* than the required extra potential $(\nu - 1)\Phi_{\text{bar}}$, so the depth is amply available. The bound $\|K\| \leq 1$ (the operator’s proven Herglotz–Nevanlinna boundedness) constrains the *form factor* acting on the baryon density, not the metric-derivative terms in T_{00} . The pure-MI deficit is therefore **structural** — which sector the enhancement lives in — not a shortfall of magnitude. What remains genuinely uncomputed, and would sharpen horn C from “structurally under-lenses” to a fully evaluated magnitude, is the deep-MOND curved-space variation $\delta S_{\text{matter}} / \delta g^{\mu\nu}$ keeping all $K'(\square_u)$ terms — which would also settle the one loophole horn A’s argument sets aside, an anisotropic-stress (gravitational slip) contribution that enhances lensing without touching the dynamical potential. This note does not claim that computation; it claims the structural trilemma, which stands independent of it.

5. Honest ledger

Established (verified against primary sources): the exhaustive trilemma (horns A/B/C); horn A’s realization as Deffayet–Woodard / $f(Q)$ -MOND modified gravity; horn B’s death by GW170817 and the exact conformal-invariance obstruction; the reclassification of any correct-lensing one-cone theory as modified gravity or a medium. **Withdrawn:** the “ 10^7 too weak” magnitude claim (dimensional error; the depth is available). **Open:** whether the *on-baryon* K' term mildly enhances the local T_{00} (a possible small dynamics double-count, decidable only by the full δ -connection enumeration — it does not affect the far-field under-lensing verdict, which is locked by §2.1’s support-plus-multipole argument); whether Road 2’s exponential screening passes the Cassini quadrupole. **Not claimed:** that the framework needs dark matter (it does not — both surviving roads are dark-matter-free); that the theory is closed. The program’s a_0 reframing, and Road 1’s derivation of $a_0 = c^2 \sqrt{\Lambda/32\pi}$, are untouched by this result; what is eliminated is only the specific hope that pure inertia, on one metric with no new field, could also do the lensing.

Reproducibility

Committed scripts (real_research/reviews/pure_mi_lensing_2026/): lane1_theorem.py + t3_crux.py (the weak-field stress-tensor teeth and the withdrawn-magnitude check),

lane2_nonlocal.py (the Deffayet–Woodard reduction and screening), lane3_connection.py (the trilemma and the conformal/GW obstructions), verify_audit.py. Companion records: the elastic-medium action [Zenodo 10.5281/zenodo.21303747]; program concept [10.5281/zenodo.21253644].

References

1. C. Deffayet, G. Esposito-Farèse & R. P. Woodard, *Nonlocal metric formulations of MOND with sufficient lensing*, Phys. Rev. D **84**, 124054 (2011).
2. J. B. Jiménez / D’Ambrosio, Garg & Heisenberg, *$f(Q)$ nonmetricity MOND*, Phys. Lett. B **811**, 135970 (2020).
3. M. Milgrom, *MOND — particularly as modified inertia*, Acta Phys. Polon. B **42**, 2175 (2011).
4. M. Brouwer et al., A&A **650**, A113 (2021).
5. T. Mistele & S. McGaugh, JCAP **04**, 020 (2024).
6. S. Boran, S. Desai, E. Kahya & R. Woodard, Phys. Rev. D **97**, 041501 (2018).
7. J. Bekenstein & M. Milgrom, ApJ **286**, 7 (1984).

Framework-first throughout; both a_0 footings where relevant. This note eliminates the pure-MI lensing channel and prices the two dark-matter-free roads that remain; it does not claim the program needs dark matter, and makes no theory-of-everything claim.